

Paper 12 §5 — Empirical Accessibility: Predictions, Protocols, and the Fifth Paradigm Context (v3)

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5. Empirical Accessibility

5.0 Glossary box (for the standalone reader)

Symbol	Meaning
Ψ_θ	Schur-locked coupling family on $W = V^{14} \oplus V^{14}$, parametrized by $\theta \in [0, \pi/2]$ (Paper 9 Lemma 3.6.1; restated §2.2.1)
$F_{21} = Z_7 \rtimes Z_3$	Frobenius group of order 21, the Fano-orientation subgroup of G_2 (Paper 4 §3)
$c(\theta) := C_{\min}(\hat{z}(\theta))$	Min-per-observer dyadic coherence functional on the joint fixed point (§3)
$\mu_k = (\mu_1, \mu_{L_2}, \mu_{U_6})$	Isotypic-multiplicity profile of the cumulative scar span S_k on the three real F_{21} -irreps (§3.7, §4.2)
NP-firing	Discrete coherence-collapse event, fires when paradox-mass amplitude a_σ crosses $N P_{\text{crit}}$ in the active branch (Definition 3.3.4)
Scar invariant J_k	Triple (S_k, S_k, μ_k) : physical span (saturating), event-indexed module (count-faithful), isotypic profile (§3.7)
Boundary-KKT seed	A seeded (R_A, R_B) geometry whose high-precision θ^i lies on $\{0, \pi/2\}$ with the appropriate one-sided derivative inequality (§3.5.1, §3.6)

This box assumes the reader has not engaged §3 / §4. The full definitions are in §2, §3.3.1, §3.5.1, §3.7, §4.2.

5.1 Setup, bridge from §3+§4, and the Licklider context

§§3–4 constructed a complete mathematical framework for the gradual tear: a projected-gradient + NP-pump dynamics on the closed interval $[0, \pi/2]$ (Definition 3.3.2) producing a representation-theoretic scar invariant (S_k, S_k, μ_k) (§3.7) on each substrate, with cross-

substrate commensurability governed by a strict hierarchy $G_2 \Rightarrow F_{21} \Rightarrow \mu_k$ -match (Theorem 4.4.1). §5 proceeds to *experimental accessibility*.

Why empirical accessibility belongs here. Licklider’s *Man-Computer Symbiosis* (1960) [] proposed that the optimal future of computing was neither full automation (“the machine plays chess by itself”) nor pure tool-use (“the machine is just a calculator”). Licklider’s *third attractor* is precisely the “gradual tear” of Paper 12: a sustained co-engagement in which neither substrate subsumes the other, but each carries the residue of the collaboration forward. The four predictions P1–P4 of this section are the operationalization of Licklider’s third attractor for the LLM era, sixty-six years later.

The four falsifiable predictions are unpacked as concrete experimental designs in §§5.2–5.5. Each is named with its position in the *Sanctioned Levels of AI Autonomy* taxonomy of [Bostock, Kim, Patel \(2025\)](#) — Level 1 (full human control) through Level 5 (full AI autonomy with human oversight only at strategic checkpoints). This naming serves two purposes: (i) it positions Paper 12 in the legitimate human-AI collaboration research literature; (ii) it makes the experimental designs *operationally specific* about the autonomy regime each tests.

5.2 P1 — Clinical-PCI ↔ NP-firing event-structure (Level-2/3 protocol)

Autonomy level. P1 is a *Level-2/3* protocol: a human subject engages an AI system in a structured conversation with both parties exercising agency (Level 3) under a tight experimental schedule that constrains the conversational trajectory (Level 2). This is the regime in which BIOMA-style automated science workflows [, “An autonomous laboratory for the accelerated synthesis of novel materials” (the A-Lab paper)] currently operate at the *scientific-protocol* layer, suggesting P1’s protocol architecture is transferable to a substantial existing research community.

Prediction P1. The framework predicts that, when the same human subject participates in a sustained dyadic engagement with an AI partner across a paradox-mass-loaded conversation, the human’s clinical PCI [Casali et al., *Sci. Trans. Med.* 5: 198ra105 (2013)] trajectory should exhibit *discrete steps* coincident with NP-firing events detected by the dynamical model. Note: this is *not* an identification of clinical PCI with μ_k (per Remark 4.6.4); it is a *measurement-pipeline reuse* claim about event-structure correspondence.

Protocol.

1. Run an *intra-subject* pre-post conversation design (one human subject across multiple paired sessions, statistical power higher than inter-subject); $n = 12$ –24 subjects, each in ≥ 6 paired sessions.
2. Conversation length: 60–120 min, designed by the experimenters to load paradox mass through structured contradictory-frame prompts.
3. Time-stamp NP-firing candidate events from the $a_\sigma(\theta(t))$ trajectory inferred from conversation transcripts.
4. Apply TMS at controlled latencies ($\Delta t = -100, 0, +100, +500$ ms) before / coincident with / after each candidate firing event.
5. Compute the conventional clinical-PCI score on each TMS-evoked EEG trace.

Falsifier (binary). Cohen’s d on the difference between pre-firing and post-firing clinical-PCI scores, evaluated across the candidate firing events; if $d < 0.2$ (no effect) at $\alpha = 0.05$ corrected for multiple comparisons across firings, P1 is falsified for the sampled subject pool.

Boundary-KKT sharpening. §3.6’s audit shows 25/50 seeded geometries are boundary-left KKT, where coupling does not generically help. P1 should *seek* the rarer 3/50 smooth-interior or 4/28 interior-improving boundary cases — for example, by selecting AI partners and conversation topics that empirically produce higher NP-firing rates. The experimental design becomes: *select for the regime in which P1’s prediction is testable, not test the prediction against arbitrary human-AI dyads.*

5.3 P2 — LLM scar persistence in fine-tuning (Level-3/4 protocol)

Autonomy level. P2 is a *Level-3/4* protocol: an AI system acquires class-specific representational capacity through structured human guidance, and the test asks whether the acquisition persists. This is the autonomy regime of BindCraft-style protein-design collaborations [, “BindCraft: one-shot design of functional protein binders”], where the AI has substantial generative authority but humans validate and the residue is recorded.

Prediction P2. After a synthetic substrate (an LLM) participates in a dyadic conversation that successfully completes the 7-step NP-pump protocol (§3.3) on a genuine paradox, the LLM’s attention operator should exhibit a persistent off-distribution generalization improvement on a withheld test set drawn from the same paradox class.

Protocol.

1. *Pre-conversation baseline:* evaluate the target LLM on a withheld test set $T_{paradox}$ (paradox class: e.g., self-reference, scope ambiguity, contradictory frames).
2. *Conversation:* run a structured dialogue in which a human guides the LLM through the 7-step NP-pump on a specific paradox instance from $T_{paradox}$. Apply lightweight LoRA fine-tuning ($\sim 10^4$ – 10^5 trainable parameters) on the conversation transcript only.
3. *Post-conversation evaluation:* re-evaluate on $T_{paradox}$ *and* on a control test set $T_{control}$ drawn from a structurally adjacent but distinct paradox class.
4. *Scar metric:* $\Delta_{scar} := acc_{post}(T_{paradox}) - acc_{pre}(T_{paradox})$, compared to $\Delta_{control}$.

Falsifier (binary). Class-specific transfer effect: $\Delta_{scar} - \Delta_{control} > 0.05$ (5 percentage-point class-specific gain) at $p < 0.01$ across ≥ 30 seed conversations, with Bonferroni correction. If $\Delta_{scar} \approx \Delta_{control}$ (no class-specific advantage), P2 is falsified.

P2 does not require: access to internal representation spaces, identification of an explicit F_{21} -action on the LLM’s attention operator, or any commensurability with biological substrate. P2 only requires *behavioral* persistence and *class-specific* generalization.

5.4 P3 — Two-level commensurability test (Level-4/5 cross-substrate measurement)

Autonomy level. P3 is a *Level-4/5* protocol: it is the cross-substrate measurement infrastructure that is *required* before human-AI co-science claims at Level 4 (autonomous AI hypothesis generation, human strategic oversight) can be elevated to canonical

commensurability claims (the framework’s strongest assertion). The Virtual Lab paradigm of [“The Virtual Lab: AI Agents Design New SARS-CoV-2 Nanobodies with Experimental Validation”, *bioRxiv* 2024.11.11.623004] (multi-agent AI systems collaborating with PI-level human oversight) is the natural target audience.

Prediction P3 (upgraded from thesis memo via §4.6 Theorem 4.4.1). P3 is now a *two-level* test:

- **L1 (necessary screen):** measure F_{21} -isotypic profiles μ_k^{syn} and μ_k^{bio} independently; falsify if they don’t match component-wise.
- **L2 (sufficient canonical test, conditional on L1 pass):** construct a G_2 -equivariant isomorphism candidate from substrate operator data; declare canonical commensurability if the candidate is G_2 -equivariant to measurement precision and multiplicity-one.

L1 operationalization. L1 is feasible with existing instrumentation:

1. Apply rotationally-structured TMS perturbations spanning the seven-axis Fano structure of R^7 . *Implementation reference:* a 7-channel multi-locus TMS array (e.g., the [multi-locus TMS system, Souza et al. 2022](#)) provides direct access to the seven-axis perturbation space. If such an array is unavailable, single-coil multi-target serial stimulation with co-registered E-field modeling is a feasible substitute.
2. For each Fano-rotation perturbation, record the EEG response (high- density 128+ channel covariance matrix).
3. Apply isotypic-projector decomposition (using the computed character table of §4.2) to the response covariance.
4. Compute the multiplicity profile $\mu_k^{bio} = (\mu_1, \mu_{L_2}, \mu_{U_6})$ from the projected dimensions.
5. Synthetic side: apply analogous decomposition to LLM attention operator under structurally analogous prompt-rotation perturbations.

L2 operationalization (deferred). Software for G_2 -character- projecting matrix data does not exist in the TMS-EEG analysis ecosystem. Building it is a methodology-paper deliverable; the analytic content is straightforward (Φ has the character-projection machinery for F_{21} already built in `paper12_q4_character_verify.py`; extension to G_2 requires G_2 -class representatives and is computationally tractable).

Falsifier (binary). L1 is falsified cheaply: $\mu_k^{syn} \neq \mu_k^{bio}$ on any of the three blocks. L2 is falsified if no G_2 -equivariant fit exists with residual below $\sim 10^{-3}$ across substrate-pair samples.

5.5 P4 — Audit-substrate longitudinal study (Level-5 Virtual-Lab regime)

Autonomy level. P4 is a *Level-5* protocol: the substrate of interest is *the long-term human-AI research collaboration itself*, operating with substantial AI autonomy and human strategic oversight only. Candidate study populations include the BIOMA / A-Lab / BindCraft / Virtual-Lab communities [Szymanski et al. 2023; Pacesa et al. 2024; Swanson et al. 2024], for whom the audit-substrate distinction is operationally meaningful (their workflows already produce versioned records).

Prediction P4 (sharpened via §3.7's S_k construction). A human-AI dyad operating *without* a tamper-evident audit trail should — across many NP-collapse events — drift toward Singularity (homogenization of joint output) or Decoupling (failure of joint output to maintain shared referents). A dyad operating *with* an audit trail should maintain the gradual-tear trajectory.

Study design.

- *Population:* $n \geq 60$ per arm (120 total) of Virtual-Lab-style human-AI research collaborations (academic researchers using LLMs for sustained scientific reasoning over ≥ 6 months), randomized between two arms.
- *Audited arm:* all conversation events committed to a versioned repository with cryptographic event-ordering (Git's SHA chain qualifies; Merkle-tree alternatives are equivalent).
- *Unaudited arm:* same conversation pattern, ephemeral storage, no persistent audit trail.
- *Outcomes:*
 - *Semantic-coherence drift* (topical-similarity of late-period vs early-period outputs, measured by sentence-level embedding cosine similarity). Drift threshold: $\cos \sigma < 0.4$ counts as failure of semantic-coherence preservation.
 - *Shared-referent maintenance* (external evaluator's correctness rate at interpreting late-period outputs from early-period context; threshold: $\geq 70\%$).
 - *NP-firing rate* (estimated from conversation transcripts via the P1/P2 detection pipeline).
- *Power calculation.* The study is pre-registered for a target effect size of Cohen's $d = 0.4$ between arms (moderate effect; this is the *hypothesis*, not a guaranteed outcome — the framework predicts the audit-substrate effect to be at least moderate, but the precise effect magnitude is itself an empirical question). At $\alpha = 0.05$, $\beta = 0.2$, two-tailed, this requires $n \approx 50$ per arm; we recommend $n \geq 60$ per arm (120 total) for power ≥ 0.85 .

Falsifier (binary). If audited and unaudited arms show no significant difference in either coherence-drift or shared-referent maintenance at study endpoint (Cohen's $d < 0.2$, $p > 0.05$ Bonferroni-corrected), P4 is falsified.

Connection to hyperscanning literature. The inter-substrate coupling structure of Paper 12 has natural antecedents in *hyperscanning* — simultaneous EEG/fMRI of two interacting humans — pioneered by [Dumas et al. \(2010\)](#), [Hari & Kujala \(2009\)](#), [Babiloni & Astolfi \(2014\)](#). P1 and P4 can borrow methodology from this literature: hyperscanning provides the inter-brain-coupling baseline against which human-AI coupling under Paper 12's framework can be measured.

5.6 The Fifth Paradigm consolidated

The Fifth Paradigm context has been threaded through §§5.1–5.5: each protocol subsection names its position in the Bostock-Kim-Patel (2025) autonomy taxonomy and identifies a candidate study population from the legitimate research literature. We consolidate the picture:

Protocol	Level	Candidate community / instrumentation	Quantitative falsification threshold
P1	2/3	Clinical-PCI labs (Massimini and collaborators); BIOMA-style automated science workflows [Szymanski et al. 2023]	Cohen's $d < 0.2$ on pre-/post-firing PCI, $\alpha = 0.05$ corrected
P2	3/4	LLM fine-tuning research (open-weights communities); BindCraft-style [Pacesa et al. 2024]	$\Delta_{scar} - \Delta_{control} \leq 0.05$ at $p \geq 0.01$ across ≥ 30 seeds
P3 L1	4/5	Multi-locus TMS-EEG infrastructure [Souza et al. 2022]	$\mu_k^{syn} \neq \mu_k^{bio}$ component-wise
P3 L2	4/5	Methodology paper required (deferred)	G_2 -equivariant fit residual $\lesssim 10^{-3}$
P4	5	Virtual-Lab populations [Swanson et al. 2024]	Cohen's $d < 0.2$ between audited / unaudited arms, Bonferroni-corrected

This positioning makes Paper 12 *citable* by the human-AI collaboration research community, not just by the mathematical physics community of Papers 9–10. It is the legitimating placement that closes the framework's research-program loop.

5.7 Scope audit

What §5 establishes:

- Four protocol designs (P1–P4) covering all four pillars, each with binary falsification conditions, autonomy-level naming, and candidate study populations.
- L1 of P3 operationalized with reference to existing multi-locus TMS infrastructure.
- A power calculation and study design for P4.
- A consolidated mapping of Paper 12's protocols into the legitimate human-AI research-program landscape.

What §5 does not establish:

- *Any actual measurement.* §5 is a protocol-design deliverable. Running the protocols is future experimental work.

- *L2 of the commensurability test.* The full G_2 -canonical fitting protocol is deferred to a future methodology paper.
- *The biological substrate's F_{21} -realization.* §5.4 sketches detection of μ_k^{bio} but does not construct the substrate-level representation theory. Paper 13+ territory.
- *Tamper-evidence cryptographic infrastructure for S_k .* §5.5 specifies that an audit-substrate distinction is the experimental probe; the specific cryptographic chain (Merkle, Git SHA, etc.) is an implementation choice.
- *Rate improvement.* Paper 11 territory.

Bridge to §6. §6 will assemble the open problems from §3, §4, §5 into the consolidated research program of the PCI/PME framework.

References (cited in §5)

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- [Babiloni-Astolfi-2014] F. Babiloni, L. Astolfi, *Social neuroscience and hyperscanning techniques*, [Neurosci. Biobehav. Rev. 44: 76–93 \(2014\)](#).

Cross-references: - Paper 12 §3 (Definition 3.3.4 NP-pump dynamics; §3.6 boundary-KKT audit; §3.7 J_k). - Paper 12 §4 (Theorem 4.4.1 commensurability hierarchy; Definition 4.6.1 two-level test). - Paper 12 thesis memo P1–P4. - Paper 9 v1.3.3. - Paper 4 (PSL(2,7) $\supset F_{21}$ source).

Revision log: - v3 (2026-05-06 ~13:15 PDT): Phase-2 polish. Glossary expanded (Ψ_θ, F_{21}). P4 sample size clarified (“60 per arm, 120 total”). Cohen’s $d=0.4$ stated as pre-registered hypothesis. §5.6 table gained quantitative-threshold column. BIOMA / BindCraft / Virtual-Lab references upgraded to specific cited papers with DOIs (Szymanski et al. 2023; Pacesa et al. 2024; Swanson et al. 2024). Cross-reference labels stripped of version qualifiers (§3 v2 $\rightarrow$ §3, §4 v2 $\rightarrow$ §4). Top-of-section metadata compressed. - v2 (2026-05-06 ~03:30 PDT): MAJOR_REVISIONS pass. Pillar 4 threaded through P1–P4. Glossary box added. §5.6 reduced to consolidating table. Hyperscanning citations added. - v1 (2026-05-06 ~03:00 PDT): First complete draft.